

10 Frequency Response Functions

In review of last time for a bit.

Remember that we can represent $x(t)$ with the Fourier series as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

The response through a LTI system is

$$y(t) = \sum_{k=-\infty}^{\infty} H(jk\omega_0) a_k e^{jk\omega_0 t}$$

Where $H(jk\omega_0)$ is the frequency response function, i.e.

$$\frac{1}{1 + j\omega}$$

OR, more clearly

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega_0 t}$$

where

$$b_k = H(jk\omega_0) a_k$$

For example:

$$x = \cos(10t)$$

$$a_1 = \frac{1}{2}$$

We know

$$b_1 = a_1 \frac{1}{1 + 10j} = 0.05e^{-1.4j}$$
$$b_2 = a_2 \frac{1}{1 - 10j} = 0.05e^{1.4j}$$

The phase

$$= \tan^{-1} \left(\frac{0}{1} \right) - \tan^{-1} \left(\frac{20}{2} \right)$$
$$\tan^{-1} \left(\frac{\text{top imaginary}}{\text{top real}} \right) - \tan^{-1} \left(\frac{\text{bot imaginary}}{\text{bot real}} \right)$$

Discrete time

if we can come up with a representation of $x[n]$ in the form

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\hat{\omega}_0 n}$$

then we have

$$y[n] = \sum_{k=-\infty}^{\infty} H(e^{jk\hat{\omega}_0}) e^{jk\hat{\omega}_0 n}$$

Or similarly,

$$\text{let } b_k = H(e^{jk\hat{\omega}_0})a_k$$
$$y[n] = \sum_{k=\langle N \rangle} b_k e^{jk\hat{\omega}_0 n}$$

For Example:

$$x[n] = \cos\left(\frac{\pi}{2}n\right). N = 4, \hat{\omega}_0 = \frac{\pi}{2}$$
$$a_1 = \frac{1}{2}, a_k = a_{k+N}$$

If

$$H(e^{j\hat{\omega}}) = \frac{1}{2}(1 + e^{-j\hat{\omega}})$$

then

$$b_1 = \frac{1}{2} \left(\frac{1}{2} (1 + e^{-j\frac{\pi}{2}}) \right) = \frac{\bar{2}}{4} e^{-j\frac{\pi}{4}}$$
$$b_2 = \frac{1}{2} \left(\frac{1}{2} (1 + e^{j\frac{\pi}{2}}) \right) = \frac{\bar{2}}{4} e^{j\frac{\pi}{4}}$$

So

$$y = \frac{\bar{2}}{2} \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right)$$